

Corbin Savings

Corbin Savings had 8 percent capital, a clean loan book, no credit problems, and was economically insolvent, and none of those four facts appeared on the same page of any report the board received.

Corbin was a \$6.4 billion savings institution in the upper Midwest, funded almost entirely by retail deposits. Its asset side was long: \$2.9 billion of thirty-year fixed-rate residential mortgages originated between 2020 and 2022 at an average rate of 3.1 percent, \$1.8 billion of agency mortgage-backed securities bought in the same period, and \$1.1 billion of Treasuries. Cash and other assets made up the rest. The weighted average duration of the asset portfolio was 5.7 years.

The liability side was short and, on paper, shorter still. Deposits were 91 percent of funding. Roughly two-thirds were checking and savings balances with no contractual maturity, which Corbin's asset-liability model treated as having a duration of about 2 years based on the historical stickiness of the balances. The remainder were certificates of deposit averaging fourteen months. The model's weighted average liability duration was 1.7 years.

Equity was 8 percent of assets.

Corbin's board received a quarterly interest-rate risk report. It showed net interest income under a plus-100-basis-point and a minus-100-basis-point scenario, both over a twelve-month horizon, and both were comfortably within policy limits. Corbin's net interest income was not very sensitive to rates over one year, because most of its assets were fixed and most of its deposits repriced slowly.

What the report did not show was the economic value of equity.

Wilhelmina Tarrant, who had joined the board in the autumn from a commercial bank, asked for it at the January meeting. She asked in a specific way: not for a scenario, but for the duration of Corbin's equity.

Her second question was whether the 2-year duration on the non-maturity deposits was an estimate or an assumption, and what it had been estimated from.

	AMOUNT	DURATION
Residential mortgages, 30-year fixed	\$2,900m	7.1
Agency mortgage-backed securities	\$1,800m	6.2
Treasuries	\$1,100m	4.4
Cash and other	\$600m	0.0
Total assets	\$6,400m	5.7
Non-maturity deposits	\$3,880m	2.0
Certificates of deposit	\$1,940m	1.2
Other liabilities	\$68m	0.5
Total liabilities	\$5,888m	1.7
Equity	\$512m	?

EXHIBIT 2 — TWO BONDS, FOR REFERENCE

Both priced to a 5.00 percent yield, both maturing in ten years.

	8% COUPON	ZERO COUPON
Price	123.384	61.027
Macaulay duration	7.393	10.000
Modified duration	7.212	9.756

ASSIGNMENT

- 1 Using Exhibit 2, compute the price change predicted by modified duration for a 100 basis point rise in yield, and the actual price change. Explain the difference and its sign.
- 2 The two bonds in Exhibit 2 have identical maturities and very different durations. Explain why, in terms of when the money arrives.
- 3 Compute the duration of Corbin's equity from Exhibit 1. State the formula you used.
- 4 What happens to the economic value of Corbin's equity if rates rise 100 basis points? 200?

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Tarrant's second question. If the true duration of non-maturity deposits is 0.7 years rather than 2.0, recompute your answer to question 3. How would you estimate that number, and what would make the estimate fail exactly when it matters?